

This question paper contains 4 printed pages]

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S. No. of Question Paper : 2250

Unique Paper Code : 32351401 IC

Name of the Paper : Partial Differential Equations

Name of the Course : B.Sc. (Hons.) Mathematics

Semester : IV

Duration : 3 Hours

Maximum Marks : 75

(Write your Roll No. on the top immediately on receipt of this question paper.)

All questions are compulsory.

Section-I

I. Attempt any *three* parts out of the following :

(a) Determine the integral surfaces of the equation :

$$x(y^2 + u)u_x - y(x^2 + u)u_y = (x^2 - y^2)u$$

with the data $x + y = 0$, $u = 1$. 6

(b) Apply the method of separation of variables to solve the initial-value problem :

$$x^2 u_{xy} + 9y^2 u = 0, \quad u(x, 0) = \exp\left(\frac{1}{x}\right). \quad 6$$

(c) Reduce the following equation into canonical form and find the general solution :

$$u_x + u_y = u. \quad 6$$

P.T.O.

- (d) Solve the initial-value problem :

$$u_t + uu_x = 0$$

with the initial curve $x = \frac{\tau^2}{2}, t = \tau, u = \tau$. 6

Section-II

2. Attempt any *one* part out of the following :

- (a) Show that the equation of motion of a vibrating string is :

$$u_{tt} = c^2 u_{xx}, \text{ where } c^2 = T/\rho. \quad 6$$

- (b) Derive the wave equation of a string :

$$u_{tt} + au_t + bu = c^2 u_{xx},$$

where the damping force is proportional to the velocity, the restoring force is proportional to the displacement of a string, and a and b are constants. 6

3. Attempt any *two* parts out of the following :

- (a) Determine the general solution of the equation :

$$4u_{xx} + 5u_{xy} + u_{yy} + u_x + u_y = 2$$

by reducing it into canonical form. 7

- (b) Transform the equation to the form $v_{\xi\eta} = cv$, $c = \text{constant}$,

$$u_{xx} - u_{yy} + 3u_x - 2u_y + u = 0$$

by introducing the new variable $v = ue^{-(a\xi + b\eta)}$, where a and b are undetermined coefficients. 7

- (c) Classify the equation and reduce it to canonical form :

$$y^2 u_{xx} + 2xy u_{xy} + 2x^2 u_{yy} + xu_x = 0. \quad 7$$

Section-III

4. Attempt any *three* parts out of the following :

- (a) Determine the solution of the initial boundary-value problem : 7

$$\begin{aligned} u_{tt} &= 4u_{xx}, & 0 < x < \infty, t > 0, \\ u(x, 0) &= x^4, & 0 \leq x < \infty, \\ u_t(x, 0) &= 0, & 0 \leq x < \infty, \\ u(0, t) &= 0, & t \geq 0. \end{aligned}$$

- (b) Find the solution of the initial boundary-value problem : 7

$$\begin{aligned} u_{tt} &= u_{xx} & 0 < x < 2, t > 0, \\ u(x, 0) &= \sin(\pi x / 2), & 0 \leq x \leq 2, \\ u_t(x, 0) &= 0, & 0 \leq x \leq 2, \\ u(0, t) &= 0, \quad u(2, t) = 0, & t \geq 0. \end{aligned}$$

- (c) Determine the solution of the Goursat problem : 7

$$\begin{aligned} u_{tt} &= c^2 u_{xx}, \\ u(x, t) &= f(x) \text{ on } x - ct = 0 \\ u(x, t) &= g(x) \text{ on } t = t(x), \\ \text{where } f(0) &= g(0). \end{aligned}$$

- (d) Determine the solution of the initial boundary-value problem :

7

$$\begin{aligned} u_{tt} &= c^2 u_{xx}, & x > 0, \quad t > 0, \\ u(x, 0) &= f(x), & x \geq 0, \\ u_t(x, 0) &= g(x), & x \geq 0, \\ u_x(0, t) &= q(t), & t \geq 0. \end{aligned}$$

Section-IV

5. Attempt any *two* parts out of the following :

- (a) Determine the solution of the initial boundary-value problem by the method of separation of variables : 8

$$\begin{aligned} u_{tt} &= c^2 u_{xx}, & 0 < x < 1, \quad t > 0, \\ u(x, 0) &= x(1-x), & 0 \leq x \leq 1, \\ u_t(x, 0) &= 0, & 0 \leq x \leq 1, \\ u(0, t) &= u(1, t) = 0, & t \geq 0. \end{aligned}$$

- (b) Prove the uniqueness of the solution of the problem : 8

$$\begin{aligned} u_t &= k u_{xx}, & 0 < x < l, \quad t > 0, \\ u(x, 0) &= f(x), & 0 \leq x \leq l, \\ u_x(0, t) &= u_x(l, t) = 0, & t \geq 0. \end{aligned}$$

- (c) Determine the solution of the initial-boundary value problem :

8

$$\begin{aligned} u_t &= k u_{xx}, & 0 < x < \pi, \quad t > 0, \\ u(x, 0) &= \sin^2 x, & 0 \leq x \leq \pi, \\ u(0, t) &= u(\pi, t) = 0, & t \geq 0. \end{aligned}$$